Project Support Session - FoodHub

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Agenda

- Project Problem Statement and Dataset Overview
- Open Q&A
- Project Evaluation Rubric
- Project Submission Guidelines
- FAQs

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FoodHub - Business Context and Objective

The number of restaurants in New York is increasing day by day. Lots of students and busy professionals rely on those restaurants due to their hectic lifestyles. An online food delivery service is a great option for them. It provides them with good food from their favorite restaurants.

A food aggregator company FoodHub, offers access to multiple restaurants through a single smartphone app. The app allows the restaurants to receive a direct online order from a customer. The app assigns a delivery person from the company to pick up the order after it is confirmed by the restaurant. The delivery person then uses the map to reach the restaurant and waits for the food package. Once the food package is handed over to the delivery person, he/she confirm the pick-up in the app and travel to the customer's location to deliver the food.

The delivery person confirms the drop-off in the app after delivering the food package to the customer. The customer can rate the order in the app. The food aggregator earns money by collecting a fixed margin of the delivery order from the restaurants.

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FoodHub - Business Context and Objective

The food aggregator company has stored the data of the different orders made by the registered customers in its online portal. They want to analyze the data to get a fair idea about the demand for different restaurants, which will help them enhance their customer experience.

Suppose you are hired as a Data Scientist in this company, and the Data Science team has shared some of the key questions that need to be answered. Perform the data analysis to find answers to these questions that will help the company improve its business.

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FoodHub - Data Dictionary

- **The data contains different features related to a food order. The detailed data dictionary is given below.**

Data	Data Description
order_id	Unique ID of the order
customer_id	ID of the customer who ordered the food
restaurant_name	Name of the restaurant
cuisine_type	Cuisine ordered by the customer
cost	Cost of the order
day_of_the_week	Indicates whether the order is placed on a weekday or weekend (The weekday is from Monday to Friday and the weekend is Saturday and Sunday)

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FoodHub - Data Dictionary

Data	Data Description
rating	Rating given by the customer out of 5
food_preparation_time	Time (in minutes) taken by the restaurant to prepare the food. This is calculated by taking the difference between the timestamps of the restaurant's order confirmation and the delivery person's pick-up confirmation.
delivery_time	Time (in minutes) taken by the delivery person to deliver the food package. This is calculated by taking the difference between the timestamps of the delivery person's pick-up confirmation and drop-off information

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Open Q&A

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FoodHub - Evaluation Rubric

Section	Points
Understanding the Structure of the Data	5
Univariate Data Analysis	15
Bivariate/Multivariate Data Analysis	20
Quality & Use of Visualisations	6
Actionable Insights & Recommendations	6
Presentation/Notebook - Overall Quality	8

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FoodHub - Evaluation Rubric

Section 1: Understanding the Structure of the Data

What to Cover?	Guiding Questions and Considerations
Overview of dataset shape and datatypes	How many rows and columns are present? Are the datatypes appropriate?
Checking the Statistical summary of the numerical variables and missing values	Do the summary statistics highlight any unusual values or distributions? Are there missing values, and how significant are they?
Answer all the key questions asked in this section	Do the ratings for non-rated orders need to be imputed? What information can you get from the statistical summary of preparation and delivery times?

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FoodHub - Evaluation Rubric

Section 2: Univariate Data Analysis

What to Cover?	Guiding Questions and Considerations
Explore distributions of numerical and categorical variables	Do the variables follow normal, skewed, or unusual distributions? Are there outliers or extreme values worth noting?
Provide key observations about central tendency, spread, and frequency patterns	Check the central tendency of the numerical variables and how are they spread. For categorical variables, check the frequency distribution for categories
Answer all the key questions asked in this section	How to understand the spread of order costs or delivery times? How to calculate percentages for conditions like “orders above $20”?

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FoodHub - Evaluation Rubric

Section 3: Bivariate/Multivariate Data Analysis

What to Cover?	Guiding Questions and Considerations
Explore relationships between key variables	Consider all combinations of variables - numerical–numerical, categorical–numerical, categorical–categorical Which variables show strong correlations or meaningful patterns with each other?
Provide observations on correlations, trends, and associations	Are there interactions or combined effects between variables worth highlighting?
Answer all the key questions asked in this section	How do you calculate the total time taken to deliver food? How to apply business rules to calculate revenue?

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FoodHub - Evaluation Rubric

Section 4: Quality & Use of Visualisations

What to Cover?	Guiding Questions and Considerations
Use appropriate plots (histograms, bar charts, boxplots, scatterplots, etc.)	Is the chosen visualization suitable for the variable type and analysis? Do the plots highlight key patterns, trends, or anomalies effectively?
Provide clear and concise observations from each visualization	It's important to provide clear and concise observations for all tables and visualizations

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FoodHub - Evaluation Rubric

Section 5: Actionable Insights and Recommendations

What to Cover?	Guiding Questions and Considerations
Conclude with the key takeaways for the business	What are the key things that we learned from the analysis conducted that's useful for the business? Logical next steps are a key inclusion

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FoodHub - Evaluation Rubric

Section 6: Presentation/Notebook - Overall Quality

Low-code Considerations	Full-code Considerations
Clear structure, flow, and visual appeal - everything sits well in a story	Clear structure and flow
Crispness - Focus on key points	Well-commented and executed code
All sections of the rubric included, even if in an appendix	All sections of the rubric included, even if in an appendix

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FoodHub - Low-Code Version

For learners who aspire to be in managerial roles in the future, focusing on solution review, interpretation, recommendations, and communication with business stakeholders

Download the dataset and the *Template Notebook: Low-code Version* (this is a template notebook)

Fill in the blanks in the notebook to complete and execute the code to solve the questions and perform all the tasks as per the grading rubric

Once the notebook is completely executed and necessary outputs obtained, a business presentation (using Microsoft PowerPoint, Google Slides, etc.) has to be created

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FoodHub - Low-Code Version

The presentation should contain observations, insights, and recommendations for the business problem

The presentation template provided can be referred to as a sample

Once the presentation is complete, convert the presentation to .pdf format

The presentation should be submitted as a PDF file (.pdf) and NOT as a .pptx file

Please make sure that all the sections mentioned in the grading rubric have been covered in the submission

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FoodHub - Full-Code Version

For learners who aspire to be in hands-on coding roles in the future, focusing on building solution codes from scratch

Download the dataset and the *Template Notebook: Full-code Version* (this is a template notebook containing high-level steps to perform and insight-based questions)

Write necessary code to solve the questions and perform all the tasks as per the grading rubric

Clearly write down observations, insights, and recommendations for the business problem based on the analysis performed

Once the notebook is complete, download it as a **.ipynb** file and convert it to a **.html** file

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FoodHub - Full-Code Version

The notebook should be submitted as an HTML file (.html) and NOT as a notebook file (.ipynb)

The conversion can be done via one of the following ways:

Jupyter Notebook: ***Download as HTML*** from the ***File*** menu
Google Colab: Use free online tools

Please make sure that all the sections mentioned in the grading rubric have been covered in the submission

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Open Q&A

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FoodHub - FAQ

Is there a way to transfer the graphs from Colab to the presentation without it looking blurry?

There are **multiple ways** to transfer the graphs.

1. Use the following line of code just after the visualization code:

   ```
   plt.savefig("output.jpg", bbox_inches='tight')
   ```

   For example:

   ```
   sns.histplot(data=data, x='column')
   plt.savefig("output.jpg", bbox_inches='tight')
   ```

2. Use the **snipping tool** to snip the visual plot from the Python Notebook and paste the snip into PPT.
3. **Right-click** on the image and click on copy and paste the copied plot into the PPT or document.

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FoodHub - FAQ

What should the presentation look like?

The presentation should be made keeping in mind that the audience will be the **Data Science team of an organization** and should focus on explaining the **key takeaways** in an easy-to-understand manner. The visualizations used should be **readable** and it is better to **avoid showing codes** unless they are the focal point of your presentation.

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FoodHub - FAQ

How should I approach Question 13?

Question:

The company wants to provide a **promotional offer** in the advertisement of the restaurants. The condition to get the offer is that the restaurants must have a **rating count of more than 50** and the **average rating should be greater than 4**. Find the restaurants fulfilling the criteria to get the promotional offer.

The question requires filtering restaurants that satisfy a **combination of two conditions**. Let us break it down into smaller subproblems to make the resolution easier.

- Clean the **ratings** column and keep only **valid numeric values**.
- Count how many ratings each restaurant has and filter those above the **required threshold**.
- For the filtered restaurants, calculate the **average rating** and keep only those above the **cutoff**.
- Get the **final list of restaurants** that meet both conditions.

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FoodHub - FAQ

In Question 14, do we need to add the cost of the orders while calculating total revenue?

Question:

The company charges the restaurant **25% on the orders having cost greater than 20 dollars** and **15% on the orders having cost greater than 5 dollars**. Find the **net revenue** generated by the company across all orders.

The **total revenue** should be calculated based on the applied surcharges only.

- Review the **cost of each order** and apply the **correct commission rate** based on the given conditions.
- Calculate the **revenue for each order** individually using these conditions.
- **Sum up the revenues** from all orders to get the company's **total net revenue**.

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Power Ahead!

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